```
import torch
from diffusers import AutoPipelineForText2Image
from PIL import Image
import os

pipe = AutoPipelineForText2Image.from_pretrained(
    "stabilityai/sd-turbo",
    torch_dtype=torch.float16
)

device = "cuda" if torch.cuda.is_available() else "cpu"
pipe = pipe.to(device)

prompts = [
    "Chest X-ray, AP view, elderly female, clear lung fields, no inf
    "Chest X-ray, PA view, adult, focal lobar consolidation in right
    "Chest X-ray, PA view, focal opacity in right middle lobe, radio
    "Chest X-ray, PA view, blunting of costophrenic angle, pleural e
    "Chest X-ray, PA view, cardiomegaly with enlarged cardiac silhou
    "Chest X-ray, AP view, endotracheal tube and central venous cath
    "Chest X-ray, PA view, motion artifact causing blurred ribs and 
    "Chest X-ray, AP view, supine position, portable ICU radiograph"
    "Chest X-ray, portable AP imaging, ICU setting, low-resolution c
    "Chest X-ray, PA view, cardiomegaly with enlarged cardiac silhou
]

output_dir = "Medical_dataset"
os.makedirs(output_dir, exist_ok=True)

for idx, prompt in enumerate(prompts):
    image = pipe(
        prompt=prompt,
        num_inference_steps=4,
        guidance_scale=0.0
    ).images[0]

    image.save(f"{output_dir}/image_{idx+1}.png")

print("Synthetic dataset generated successfully!")
```

```
Flax classes are deprecated and will be removed in Diffusers v1.0.0.
Flax classes are deprecated and will be removed in Diffusers v1.0.0.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_valid
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your se
```

```
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
  warnings.warn(
```

model_index.json: 100% 616/616 [00:00<00:00, 66.2kB/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Ple
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthe
```

Download complete: 5.16G/? [04:08<00:00, 137MB/s]

Fetching 12 files: 100% 12/12 [00:54<00:00, 4.66s/it]

Loading pipeline components...: 100% 5/5 [00:24<00:00, 7.46s/it]

Loading weights: 100% 372/372 [00:04<00:00, 91.48it/s, Materializing param=text_model.final_lay

```
You have disabled the safety checker for <class 'diffusers.pipelines
```

100% 4/4 [00:01<00:00, 2.88it/s]

100% 4/4 [00:00<00:00, 15.52it/s]

100% 4/4 [00:00<00:00, 17.11it/s]

100% 4/4 [00:00<00:00, 17.52it/s]

100% 4/4 [00:00<00:00, 17.29it/s]

100% 4/4 [00:00<00:00, 17.38it/s]

100% 4/4 [00:00<00:00, 17.41it/s]

100% 4/4 [00:00<00:00, 17.24it/s]

100% 4/4 [00:00<00:00, 15.63it/s]

100% 4/4 [00:00<00:00, 15.17it/s]

```
Synthetic dataset generated successfully!
```

```python
import os
from PIL import Image
import matplotlib.pyplot as plt

folder = "/content/Medical_dataset/"

for file in os.listdir(folder):
    if file.endswith(".png"):
        img = Image.open(folder + file)
        plt.figure(figsize=(4,4))
        plt.imshow(img)
        plt.axis('off')
        plt.title(file)
```

```
plt.show()
```

image_9.png

image_1.png

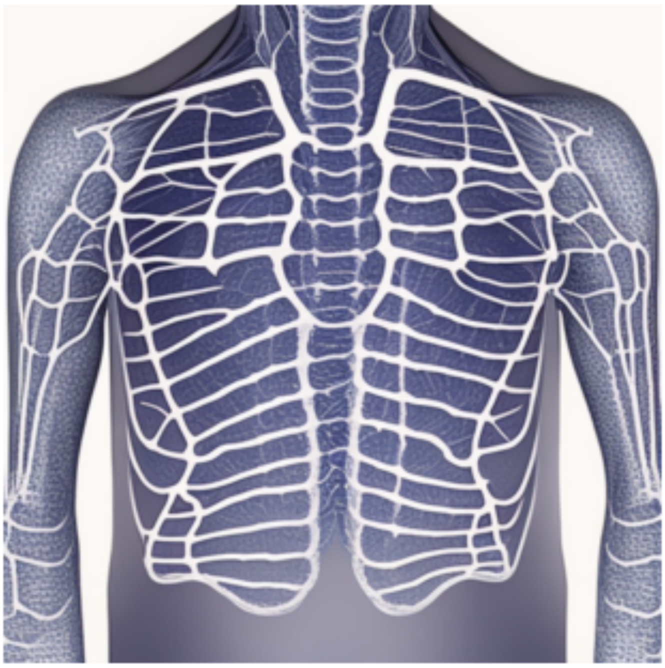
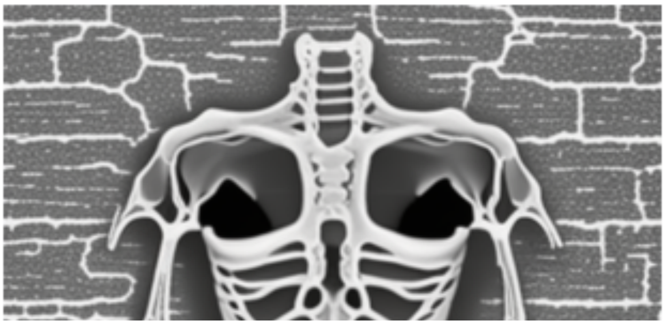
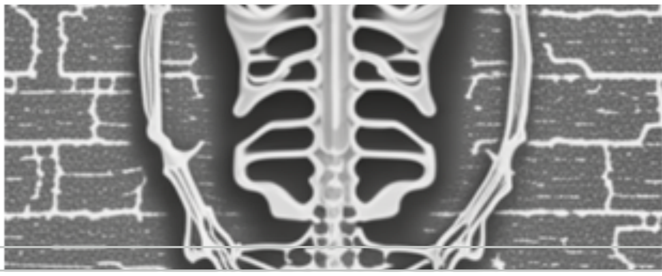


image_2.png





```
pip install torchxrayvision
```

```
Collecting torchxrayvision
  Downloading torchxrayvision-1.4.0-py3-none-any.whl.metadata (18 kE
Requirement already satisfied: torch>=1 in /usr/local/lib/python3.12
Requirement already satisfied: torchvision>=0.5 in /usr/local/lib/py
Requirement already satisfied: scikit-image>=0.16 in /usr/local/lib/
Requirement already satisfied: tqdm>=4 in /usr/local/lib/python3.12/
Requirement already satisfied: numpy>=1 in /usr/local/lib/python3.12
Requirement already satisfied: pandas>=1 in /usr/local/lib/python3.1
Requirement already satisfied: requests>=1 in /usr/local/lib/python3
Requirement already satisfied: pillow>=5.3.0 in /usr/local/lib/pytho
Requirement already satisfied: imageio in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/pythor
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/pyth
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/loca
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/pythor
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/
Requirement already satisfied: scipy>=1.11.4 in /usr/local/lib/pytho
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/pytho
Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib
Requirement already satisfied: packaging>=21 in /usr/local/lib/pytho
Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/py
Requirement already satisfied: filelock in /usr/local/lib/python3.12
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/loc
Requirement already satisfied: setuptools in /usr/local/lib/python3.
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/pytho
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/d
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/pytho
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/l
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /u
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in 
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /u
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /u
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /u
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/loca
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/lc
```

image 3.png

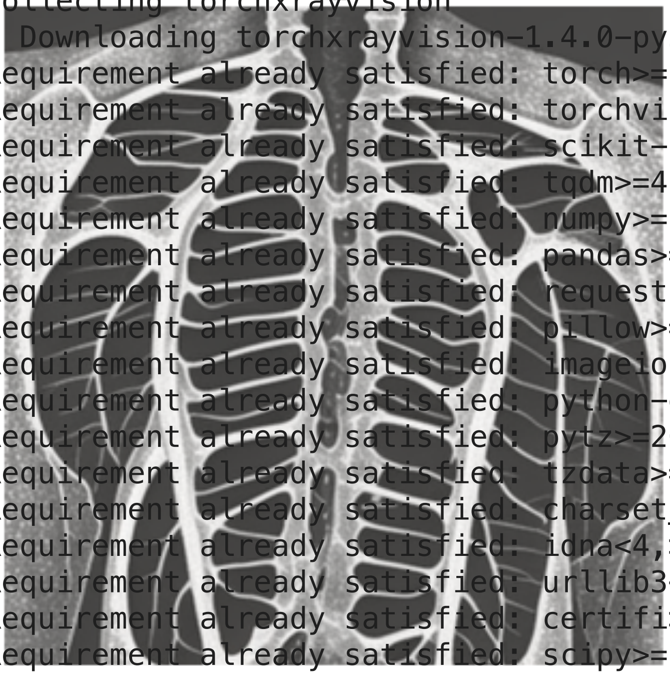


image_6.png

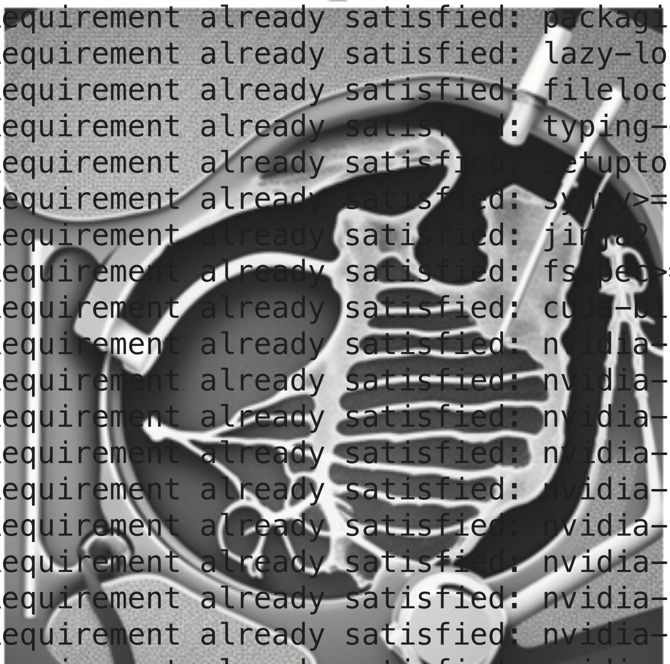
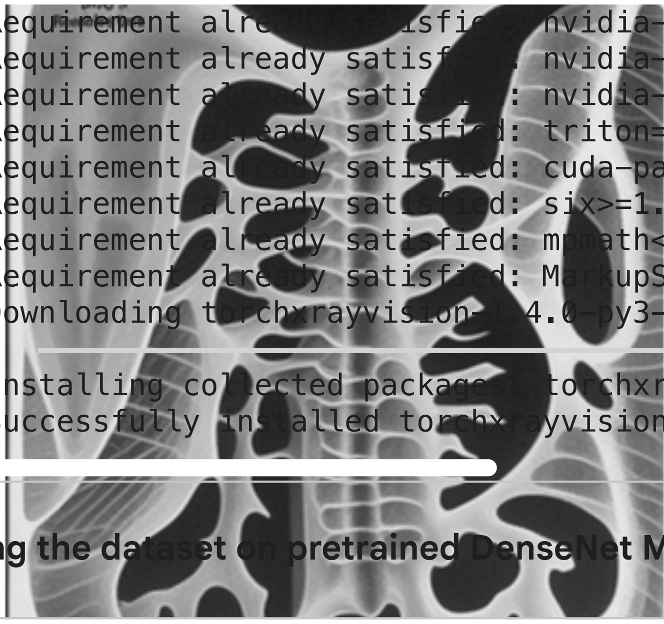


image 4.png

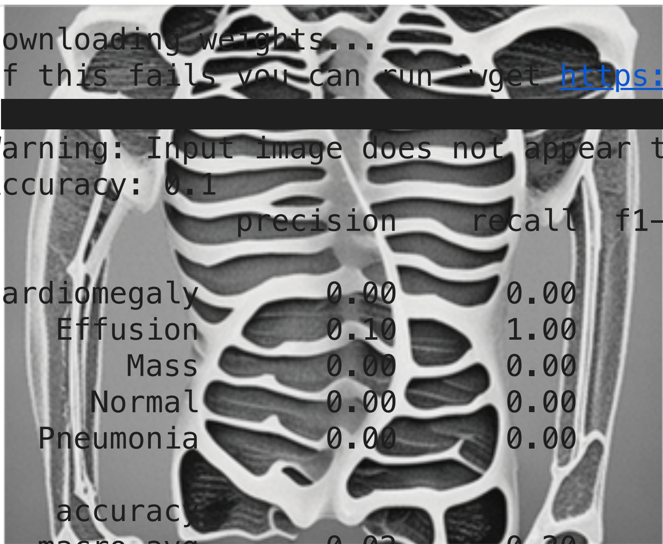


```
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/loc
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /us
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/pytho
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/li
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/pyt
Downloading torchxrayvision-1.4.0-py3-none-any.whl (29.0 MB)
                                          29.0/29.0 MB 21.0 MB/s e
Installing collected packages: torchxrayvision
Successfully installed torchxrayvision-1.4.0
```

Testing the dataset on pretrained DenseNet Model

```
import torch
import torchvision.transforms as transforms
import torchxrayvision as xrv
from PIL import Image
import os
from sklearn.metrics import accuracy_score, classification_report

model = xrv.models.DenseNet(weights="all")
model.eval()

transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor()
])

data_path = "/content/Medical_dataset"

y_true = [
    "Normal",
    "Pneumonia",
    "Mass",
    "Effusion",
    "Cardiomegaly",
    "Normal",
    "Normal",
    "Normal",
    "Normal",
    "Cardiomegaly"
]

pathologies = model.pathologies
y_pred = []
```

```
for file in sorted(os.listdir(data_path)):
    img = Image.open(os.path.join(data_path, file)).convert("L")
    img = transform(img).unsqueeze(0)

    with torch.no_grad():
        output = model(img)[0]

    idx = output.argmax().item()
    predicted_label = pathologies[idx]
    y_pred.append(predicted_label)


mapping = {
    "Cardiomegaly": "Cardiomegaly",
    "Pneumonia": "Pneumonia",
    "Mass": "Mass",
    "Effusion": "Effusion",
}

y_pred_final = [mapping.get(lbl, "Normal") for lbl in y_pred]

acc = accuracy_score(y_true, y_pred_final)
print("Accuracy:", acc)

print(classification_report(y_true, y_pred_final))
```



```
Downloading weights...
If this fails you can run `wget https://github.com/mlmed/torchxrayvi
[[redacted]]
Warning: Input image does not appear to be normalized correctly. The
Accuracy: 0.1
              precision    recall  f1-score   support

Cardiomegaly       0.00      0.00      0.00         2
    Effusion       0.10      1.00      0.18         1
        Mass       0.00      0.00      0.00         1
      Normal       0.00      0.00      0.00         5
   Pneumonia       0.00      0.00      0.00         1

    accuracy                           0.10        10
   macro avg       0.02      0.20      0.04        10
weighted avg       0.01      0.10      0.02        10
```

```
print("Accuracy of the dataset that created through hugging face or
```

```
Accuracy of the dataset that created through hugging face on a pretr
```

Testing the Real Dataset on the Pretrained model DenseNet

```
!mkdir ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
!pip install kaggle
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or di
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/d
Requirement already satisfied: bleach in /usr/local/lib/python3.12/d
Requirement already satisfied: certifi>=14.05.14 in /usr/local/lib/p
Requirement already satisfied: charset-normalizer in /usr/local/lib/
Requirement already satisfied: idna in /usr/local/lib/python3.12/dis
Requirement already satisfied: protobuf in /usr/local/lib/python3.12
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/
Requirement already satisfied: python-slugify in /usr/local/lib/pyth
Requirement already satisfied: requests in /usr/local/lib/python3.12
Requirement already satisfied: setuptools>=21.0.0 in /usr/local/lib/
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.1
Requirement already satisfied: text-unidecode in /usr/local/lib/pyth
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dis
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/pyt
Requirement already satisfied: webencodings in /usr/local/lib/python
```

```
!kaggle datasets download -d paultimothymooney/chest-xray-pneumonia
```

```
Traceback (most recent call last):
  File "/usr/local/bin/kaggle", line 10, in <module>
    sys.exit(main())
             ^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/cli.py", line
    out = args.func(**command_args)
          ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/api/kaggle_ap
    with self.build_kaggle_client() as kaggle:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/api/kaggle_ap
    username=self.config_values['username'],
             ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^
KeyError: 'username'
```

```
!unzip /content/chest_xray.zip
```

Streaming output truncated to the last 5000 lines.

```
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person23_bacter
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1728_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person420_bacte
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person978_bacte
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person808_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person325_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person46_bacter
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1361_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person385_bacte
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person868_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1657_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1228_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person851_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person348_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person417_bacte
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1369_viru
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1723_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person8_bacteri
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1231_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1182_viru
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person327_virus_
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1739_bact
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person567_bacte
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1019_viru
  inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
  inflating: chest_xray/chest_xray/train/PNEUMONIA/person1060_viru
```

```
inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
inflating: chest_xray/chest_xray/train/PNEUMONIA/person435_bacte
inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
inflating: chest_xray/chest_xray/train/PNEUMONIA/person351_virus_
inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
inflating: chest_xray/chest_xray/train/PNEUMONIA/person531_bacte
inflating: __MACOSX/chest_xray/chest_xray/train/PNEUMONIA/._pers
inflating: chest_xray/chest_xray/train/PNEUMONIA/person1682_bact
```

```
import torch
import torchvision.transforms as transforms
import torchvision.datasets as datasets
from sklearn.metrics import accuracy_score
import torchxrayvision as xrv

model = xrv.models.DenseNet(weights="all")
model.eval()

transform = transforms.Compose([
    transforms.Grayscale(),
    transforms.Resize((224,224)),
    transforms.ToTensor()
])

test_path = "/content/chest_xray/test"
test_data = datasets.ImageFolder(test_path, transform=transform)
test_loader = torch.utils.data.DataLoader(test_data, batch_size=16,

idx_to_label = {v:k for k,v in test_data.class_to_idx.items()}

pathologies = model.pathologies

y_true = []
y_pred = []

for imgs, labels in test_loader:
    with torch.no_grad():
        outputs = model(imgs)

    preds = outputs.argmax(1)

    for p in preds:
        predicted_pathology = pathologies[p]

        if predicted_pathology in ["Pneumonia", "Infiltration"]:
            y_pred.append("PNEUMONIA")
        else:
            y_pred.append("NORMAL")

    for l in labels:
        y_true.append(idx_to_label[int(l)].upper())
```

```
acc = accuracy_score(y_true, y_pred)
print("Test Accuracy:", acc*100)
```

```
Test Accuracy: 37.5
```

```
!kaggle datasets download -d paultimothymooney/chest-xray-pneumonia
!unzip ./data/chest-xray-pneumonia.zip -d ./data/
```

```
Traceback (most recent call last):
  File "/usr/local/bin/kaggle", line 10, in <module>
    sys.exit(main())
             ^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/cli.py", line
    out = args.func(**command_args)
          ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/api/kaggle_ap
    with self.build_kaggle_client() as kaggle:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-packages/kaggle/api/kaggle_ap
    username=self.config_values['username'],
             ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^
KeyError: 'username'
unzip:  cannot find or open ./data/chest-xray-pneumonia.zip, ./data/
```

```
import torch
import torchvision.transforms as transforms
import torchvision.datasets as datasets
from sklearn.metrics import accuracy_score
import torchxrayvision as xrv

model = xrv.models.DenseNet(weights="all")
model.eval()

transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.Grayscale(),
    transforms.ToTensor()
])

test_data = datasets.ImageFolder("/content/chest_xray/test", transf
test_loader = torch.utils.data.DataLoader(test_data, batch_size=16,

idx_to_label = {v:k for k,v in test_data.class_to_idx.items()}
pathologies = model.pathologies

y_true, y_pred = [], []

for imgs, labels in test_loader:
    with torch.no_grad():
        outputs = model(imgs)

    preds = outputs.argmax(1)

    for p in preds:
        pred = pathologies[int(p)]
        if pred in ["Pneumonia","Infiltration"]:
            y_pred.append("PNEUMONIA")
        else:
            y_pred.append("NORMAL")

    for l in labels:
        y_true.append(idx_to_label[int(l)].upper())

acc = accuracy_score(y_true, y_pred)
print("Accuracy on small real dataset:", acc*100)
```

```
Accuracy on small real dataset: 37.5
```